

Quantifying Epistemic Uncertainty in Diffusion Models for Chest Radiography

*A study of pixel-wise generative variance as a signal for
pathology localisation and data curation*



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Declaration of Originality

I declare that this thesis is my own work and that all material drawn from other sources has been properly acknowledged. This work has not been submitted for any other degree at this or any other institution.

Use of generative AI tools(GenAI) in the preparation of this work is disclosed as follows:

- **Code writing and programming support:** GenAI was used to help write, debug, explain, and refine the Python code developed for this research. The author critically reviewed the suggestions, adapted them where necessary, and tested them. Responsibility for the final implementation, the methodological decisions, the execution of the code, and the interpretation of all computational results remained with the author.
- **Academic resource and literature searching:** GenAI served as an auxiliary tool for identifying potentially relevant academic literature, concepts, datasets, methodologies, and other research resources. No reference suggested by GenAI was accepted on the basis of the generated output alone. The author independently located and verified each source against appropriate scholarly or authoritative material, and assessed its relevance to the claims made in this thesis.
- **Report writing and evaluation support:** GenAI helped refine language, organise ideas, correct grammar, improve clarity, and evaluate draft sections of the thesis. Its suggestions were treated as editorial assistance, not as authoritative academic content. The final text, arguments, interpretations, conclusions, and presentation of findings were reviewed and determined by the author.

GenAI was not used as a substitute for independent research, critical analysis, or academic judgement.

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The experiments ran on publicly available GPU compute with session-limited runtime and limited storage. Those constraints shaped the study more than any single methodological decision did, and the choice to spend the available budget on replicating one comparison rather than extending the class set is the reason this thesis makes the argument it does.

Finally, thanks to my family, professors and peers for their support through this degree.

Abstract

Diffusion models can generate chest radiographs that are hard to tell from real ones by inspection, but they give no account of which regions of an output are reliably reconstructed and which are effectively invented. This thesis asks whether a diffusion model’s own sampling stochasticity can supply that account. A real radiograph is inverted to a fixed latent anchor by deterministic DDIM inversion and reconstructed repeatedly under stochastic sampling, so the per-pixel variance across the resulting ensemble forms an uncertainty map tied to one specific input image rather than to the generator in general. The maps are evaluated on VinDr-CXR across five pathology classes against radiologist bounding boxes on a held-out test split, to test spatial agreement. They also serve, functionally, as a criterion for selecting synthetic augmentation data. Spatial agreement exceeds chance but is modest and varies more across classes than the uncertainty on any single class does; Cardiomegaly localises worst, not because its texture is suppressed but because it is a global size judgement rather than a local appearance abnormality, so its box encloses tissue that gives a variance map nothing to concentrate on. Reducing sampling stochasticity strips out much of the raw variance while leaving localisation intact or slightly improved – evidence that a large share of the signal carries no spatial information. Selecting synthetic images by low variance conferred no advantage over a size-matched random draw, in any class or at any threshold tested. A pre-registered five-seed replication then found training variability (SD 0.038) exceeding test-sampling variability (0.028), forcing withdrawal of a single-run -0.084 AUROC result that had survived correction for multiple comparisons. The methodological upshot: a seed-level standard deviation belongs alongside any single-run bootstrap interval, and it costs little next to the price of a retracted claim.

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Chapter 1

Introduction

1.1 Motivation

Generative models now produce chest radiographs that are difficult to distinguish from real ones by visual inspection. Generated radiographs have been proposed as a solution for two persistent problems in clinical machine learning: the lack of annotated examples of rare findings, and the legal and ethical constraints on sharing patient images between institutions.

But, visual plausibility does not prove correctness. When a diffusion model is asked to render a pneumothorax, it produces an image judged realistic by how well it matches its training distribution, not according to whether or not the finding it renders is anatomically coherent. Where the model is confident and where it is guessing are not visible in the output, and the failure is therefore silent: a classifier trained on a subset of synthetic images cannot discount the areas where the generator was not reliable; a reader reading a review of a In a synthetic example there is no way to tell which structures are to be trusted. In discriminative medical imaging, uncertainty quantification is the standard response to this problem, and calibrated confidence is widely argued to be a precondition for clinical deployment. The equivalent machinery on the generative side is far less developed. Most work on synthetic medical imaging reports distributional metrics computed over whole datasets, which say nothing about any individual pixel of any individual sample.

1.2 Problem Statement

We ask whether a diffusion model’s own sampling behaviour can yield a spatially resolved account of its uncertainty, and whether that output is useful.

Reverse diffusion is stochastic i.e. running it twice from the same starting point results in two different images, because noise is injected at each denoising step. Where the model is strongly constrained by what it has learned, those images agree; where

it is not, they diverge. Repeating the process and measuring per-pixel disagreement across the ensemble gives a map of where the image generated is stable.

If each reconstruction begins from an independently sampled latent, the resulting variance conflates the arbitrariness of the starting point with the instability of the reverse trajectory. We therefore invert a real radiograph to a fixed latent and holds it constant across the ensemble. Every reconstruction then begins from the same point and differs only in the noise injected during sampling, so the remaining variance is attributable to the reverse process applied to one specific real image.

Creating such a map is not an outcome in itself; the map must correspond to something. Correspondence is tested in two independent ways. Spatially: if generative variance carries clinical information, high-variance regions should coincide with regions radiologists have marked abnormal. Functionally: if variance identifies which synthetic images are trustworthy, selecting low-variance reconstructions as augmentation data should outperform selecting the same number at random.

1.3 Research Questions

1. **RQ1 (Localisation).** Does pixel-wise variance across inversion-anchored ensembles agree spatially with expert annotations, and does that agreement vary across pathology classes?
2. **RQ2 (Dose-response).** How does the variance signal respond to changes in the sampling configuration that produces it, and does reducing variance degrade its spatial agreement?
3. **RQ3 (Comparison).** How does generative uncertainty compare with classifier-derived uncertainty and supervised saliency, evaluated on images no model in the comparison has seen?
4. **RQ4 (Downstream utility).** Does selecting synthetic images having low generative variance improve downstream classification relative to selecting the same number at random?

RQ1 and RQ2 concern the signal itself, RQ3 places it against alternatives, and RQ4 asks whether it is useful. Each is answered in a dedicated section of Chapters 5 and 6, in this order.

1.4 Contributions

1. **An inversion-anchored uncertainty estimator** that isolates reverse-process stochasticity from latent sampling, making the resulting variance attributable to a specific input image.

2. **A spatial evaluation against expert annotations** across five pathology classes on a held-out split, using per-image rather than pixel-pooled statistical inference.
3. **A dose–response analysis of the sampler.** Two configuration approaches reduce variance magnitude substantially while leaving spatial agreement intact, indicating that much of the raw signal carries no localisation information.
4. **A controlled test of uncertainty-based curation** against a size-matched random control, separating the value of the selection criterion from the value of added data — a distinction frequently collapsed in the augmentation literature.
5. **A replication protocol and an empirical noise floor.** A multi-seed replication shows that training-seed variability exceeds test-set sampling variability on this task, and that a single-run effect significant under bootstrap resampling does not survive replication. Bootstrap intervals therefore understate uncertainty in comparisons of this kind.

The last contribution emerged from testing a result this thesis had already obtained. The replication was specified in advance with an explicit decision rule, and its outcome required withdrawing a finding rather than confirming one. It is reported in Section 6.4 as a result in its own right.

Chapter 2

Background and Related Work

This chapter covers the mechanics of diffusion sampling and inversion, the terms of uncertainty quantification, the state of synthetic augmentation in medical imaging, and the saliency method used as a comparator.

2.1 Denoising Diffusion Models

Diffusion models[1] generate images by learning to reverse a gradual noising process. A forward process corrupts a sample x_0 over T steps by adding Gaussian noise on a fixed schedule, admitting a closed form at any step so that x_T is approximately standard Gaussian. A network $\epsilon_\theta(x_t, t)$ is trained to predict the noise component of x_t , and generation iteratively removes predicted noise from an initial sample.

2.1.1 Deterministic Sampling and the η Parameter

The denoising diffusion implicit model formulation generalises the sampling step to a family of processes indexed by a stochasticity parameter η . [song2021ddim] Writing $\bar{\alpha}_t$ for the cumulative noise schedule, the update is

$$x_{t-1} = \sqrt{\bar{\alpha}_{t-1}} \left(\frac{x_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_\theta(x_t, t)}{\sqrt{\bar{\alpha}_t}} \right) + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2} \epsilon_\theta(x_t, t) + \sigma_t z_t, \quad (2.1)$$

with $z_t \sim \mathcal{N}(0, I)$ and $\sigma_t = \eta \sqrt{(1 - \bar{\alpha}_{t-1}) / (1 - \bar{\alpha}_t)} \sqrt{1 - \bar{\alpha}_t / \bar{\alpha}_{t-1}}$.

Setting $\eta = 0$ removes the $\sigma_t z_t$ term and makes sampling deterministic, so a given x_T maps to exactly one x_0 ; $\eta = 1$ recovers stochastic ancestral sampling. This parameter is the direct control on ensemble divergence and therefore on the quantity we measure. At $\eta = 0$ the variance from a fixed latent is zero by construction. As η increases, reconstructions diverge further, and whether that additional divergence carries information is the substance of RQ2.

2.1.2 Latent Diffusion and Inversion

Running diffusion in pixel space is expensive at diagnostic resolutions. Latent diffusion models operate in the compressed space of a pretrained autoencoder, with a decoder restoring full resolution[2]. This matters for uncertainty estimation because variance computed in latent space is measured on a coarser grid than the image; the propagation choice made here is given in Section 3.3.

Generation from a randomly sampled latent produces an image unconnected to any real patient. To attach the process to a specific radiograph, the image must be mapped back into latent space. DDIM inversion[3] does this by running the deterministic recursion in reverse, integrating from x_0 towards x_T . Inversion makes the estimate input-specific rather than generic, and its fidelity affects everything downstream: variance measured around a poor inversion describes the neighbourhood of the wrong image.

2.2 Uncertainty Quantification in Deep Learning

Predictive uncertainty is usually divided into an aleatory and an epistemic component, the former corresponding to the uncertainty of the random component, and the latter to the uncertainty of the model own ignorance[4]. The difference is important when dealing with an inversion-anchored ensemble. The input image is a fixed image and the latent anchor is also fixed, and there is no variation in the ensemble. does not come from data, but this is because the reverse process is under determined. At that time in latent space, the signal is in the epistemic side. This justifies the framing in the title, though it should be read as a claim about the dominant component rather than clean separation, since the decomposition is not exact and the autoencoder contributes reconstruction error of its own.

The typical methods for discriminative networks are approximations of a distribution over parameters. Monte Carlo dropout[5] leaves dropout active at inference and treats repeated forward passes as posterior samples. Deep ensembles train several independently initialised networks and treat their disagreement as the estimate, generally better calibrated at higher cost[6]. Both require a supervised network trained on labelled data for the task being evaluated, so their uncertainty is uncertainty about a supervised prediction. The generative estimator studied here is unsupervised, which makes it applicable where labels are unavailable but also means the two do not measure the same uncertainty. This shapes the comparison in Section 5.4.

Estimation of uncertainty for generative models is not as well established[7]. Evaluation in medical image synthesis is still dominated by aggregate distributional metrics that cannot speak to a particular pixel of a particular sample. Some work has examined output variability across seeds as a diagnostic, but its connection to clinically meaningful structure is not established. Whether reverse-process variance, anchored to

a real image by inversion, agrees with expert pathology annotations does not appear to have been tested directly, and that is the gap we occupy.

2.3 Synthetic Augmentation in Medical Imaging

Models that are trained on paired radiograph and report data, and then fine-tuned for text-conditioned diffusion models[8], can produce chest radiographs given a description of a finding, which is the practical basis for augmenting training sets with synthetic examples of underrepresented pathologies.

There is mixed evidence that this will improve the performance of the downstream system. Reported gains vary widely, and two features of the literature make it hard to read. Synthetic data is often added without a control that adds the same number of images selected by another rule, which conflates the benefit of the selection method with the benefit of more data. And comparisons are frequently reported from a single training run with intervals derived from bootstrap resampling of the test set alone; Chapter 6 shows this to be inadequate at the effect sizes typically at issue. A further concern is contamination: if images used to generate synthetic samples also appear in evaluation, performance is inflated invisibly. This is checked in both directions in the protocol of Section 4.1.2.

2.4 Weakly Supervised Localisation

Gradient-weighted class activation[9] mapping produces a spatial map of the evidence supporting a prediction by weighting convolutional feature maps according to the gradient of the class score, and is widely used to indicate which regions drove a diagnostic output.

Grad-CAM appears here as a comparator, and its status needs stating at the outset. It is not an uncertainty estimate: it identifies the pixels that most increased a class score, which is close to a direct answer to the localisation question being scored in Chapter 5, and it derives from a network trained on the labels the annotations encode. It is therefore treated throughout as a supervised reference rather than a peer baseline. Reporting it remains worthwhile, because it indicates how much localisation signal is available to a method with full label supervision.

2.5 Research Gap

Three observations define the gap. Uncertainty quantification for generative medical imaging lacks the spatially resolved, per-sample estimates that are standard on the discriminative side. Where variability in diffusion outputs has been examined, it has not been validated against expert spatial annotations, so its clinical meaning is unknown. And proposals to curate synthetic data by confidence are rarely tested against

a control isolating the selection criterion from the quantity of data added.

We address all three, and additionally establishes what magnitude of downstream effect is distinguishable from training noise in this setting.

Chapter 3

Methodology

3.1 Overview

The anchor is reconstructed on a real radiograph which is inverted to a fixed latent anchor. Stochastic sampling is performed N times, so that the ensemble is shortened to the pixel-wise variance. Both spatially, using radiologist bounding boxes, and functionally, as a selection criterion for downstream training data. Figure 3.1 shows the arrangement.

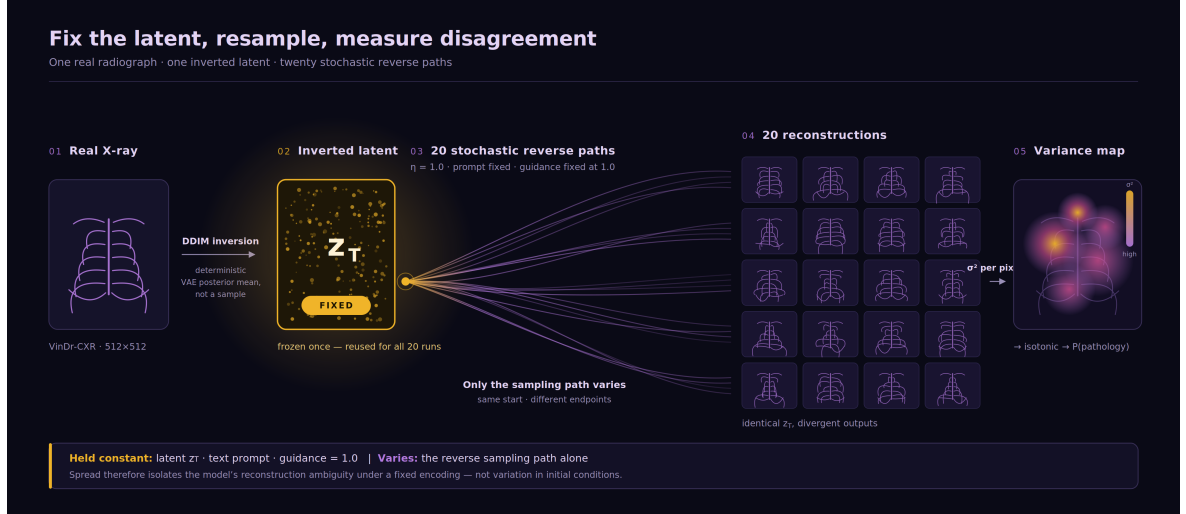


Figure 3.1: The uncertainty estimation and evaluation pipeline. A real radiograph is inverted to a fixed latent anchor and reconstructed N times under stochastic sampling. Per-pixel variance across reconstructions forms the uncertainty map, evaluated for spatial agreement with expert annotations and for utility as a selection criterion in downstream training. The two evaluation branches are independent and address different research questions.

3.2 Latent Anchoring

Each test radiograph is embedded to the autoencoder's latent space and has been inverted along with it. We use the deterministic DDIM trajectory ($\eta = 0$) to obtain a latent z_T that reconstructs. When the input is denoised deterministically, the output is

the same. This latent is calculated only once for an image, CACHED, and used for each member of an ensemble. The alternative design is the more obvious design and that’s why the choice needs justifying. If each member began from an independently drawn $z_T \sim \mathcal{N}(0, I)$, the ensemble would sample the model’s output distribution as a whole and its variance would describe global diversity rather than behaviour near a particular patient. Anchoring makes the estimate local: each reconstruction begins at the same point, so that the source of divergence is the noise that is added on the reverse pass. Two consequences follow. This estimate assumes the inversion is accurate, and should not be compared to the diversity measure calculated for independent generations.

3.3 Ensemble Construction and Variance Estimation

From the anchor, the reverse process is run N times using Equation 2.1 at fixed η . Each pass draws independent noise at every step, so the trajectories diverge from a shared origin and terminate at N distinct images after decoding. Classifier-free guidance is held at unity throughout, so conditioning does not differentially amplify divergence across images.[10]

The ensemble is reduced by the unbiased sample variance at each pixel (i, j) :

$$\sigma^2(i, j) = \frac{1}{N-1} \sum_{n=1}^N \left(x^{(n)}(i, j) - \bar{x}(i, j) \right)^2. \quad (3.1)$$

The variance is calculated in image space, not in the latent space where sampling occurs. Latent-space variance will be on a coarser grid and would make assumptions regarding the propagation of it by the decoder to the pixels; computation in image space does not make the assumption, and avoids an extra pass of the decoder.

3.3.1 Sampling Configuration Ablation

Three axes are varied independently from the reference configuration: the stochasticity parameter η , the number of denoising steps, and the ensemble size N . Varying η changes noise injected per step and is the most direct control on divergence. Varying the step count changes how many opportunities for divergence the trajectory contains. Varying N changes only how many samples the variance is estimated from, not the distribution being sampled, and therefore serves as a negative control: it should make the estimate have more noise without systematically changing what it measures.

3.4 Comparator Estimates

Two comparators are computed on the same held-out images. **Monte Carlo dropout logit variance** is obtained by evaluating a DenseNet-121 with dropout active over

repeated stochastic forward passes and retaining the per-pixel variance of the spatial logit response. This is the closest available analogue, since both it and the generative estimator measure disagreement across repeated stochastic evaluations of a fixed input. **Grad-CAM** maps come from the same classifier and are reported as a supervised reference, for the reasons given in Section 2.4.

An earlier version of this comparison evaluated all three methods on images the classifier had trained on, which advantaged the classifier-derived maps for reasons unrelated to the estimators. Moving to the held-out split removes that advantage but not all of it: both comparators come from a network trained on this label set, while the diffusion estimator has seen neither these images nor any pathology labels. The test split equalises image exposure and leaves label supervision unequal. This residual asymmetry is not correctable within the present design and is stated wherever the comparison is reported.

3.4.1 The Residual Supervision Asymmetry

An earlier version of this comparison evaluated all three methods on images that had been part of the classifier’s training set, which gave the classifier-derived maps an advantage that had nothing to do with the estimators themselves. Moving the comparison to the held-out test split removes that advantage.

It does not remove all of it. The held-out results in Section 5.4 show that the residual matters more than the part that was removed. Both comparators come from a network trained on this label set; the diffusion estimator has seen neither these images nor any pathology label. The test split equalises image exposure. It leaves label supervision unequal, and the evidence in Section 5.4 says that unequal supervision, not image exposure, is the larger source of MCD-logit’s advantage: the gap persists at large effect sizes even once every model is evaluated on images none of them has seen. This residual asymmetry is not correctable within the present design, and it is stated wherever the comparison is reported.

3.5 Evaluation Framework

3.5.1 Spatial Agreement

Ground truth is radiologist-drawn bounding boxes. For each image and class, the union of all boxes for that class is rasterised to a binary mask at map resolution, rescaling coordinates from the original acquisition resolution. Images with no annotated box for the class are excluded, since an empty or full mask admits no ranking metric. Boxes are a coarse ground truth: they are axis-aligned and enclose normal tissue alongside the finding, so a perfectly localised map cannot achieve a perfect score, and all spatial metrics measure agreement with box annotations rather than with true lesion extent.

3.5.2 Metrics

Four metrics are reported, chosen to be sensitive to different failure modes. **Pixel AUROC** ranks in-box against out-of-box pixels and is threshold-free, so it does not require the maps to be comparably scaled. **Pixel AUPRC** is reported alongside the in-box pixel fraction, since boxes occupy a small share of the image and AUPRC must be read as lift over that fraction rather than against a fixed baseline. **Dice** is computed after thresholding each map at its own percentile, so the threshold carries no label information. **Pointing hit rate** records whether the single highest-uncertainty pixel falls inside a box; it is robust to map scaling and resampling and is closest to how a reader would use the map.

Reporting all four guards against a specific failure: AUROC and pointing hit rate depend on different properties of the map, so large disagreement between them indicates a preprocessing problem rather than a difference in method quality.

3.5.3 Unit of Analysis

All spatial metrics are computed per image and aggregated across images, with confidence intervals from bootstrap resampling over images rather than pixels. This has more consequence than it appears. Pooling every pixel from every image yields on the order of 10^8 observations and correspondingly narrow intervals, but those intervals would be invalid: pixels within an image are strongly correlated and are not independent observations, and pooling allows images with large annotated regions to dominate. The image is the unit at which observations are independent and therefore the unit at which resampling is performed. Comparisons against pixel-pooled figures elsewhere should account for this difference.

3.5.4 Statistical Testing

Comparisons between methods on the same images are paired. Where three methods are compared, a Friedman test [11] is applied to the images all three cover, and pairwise Wilcoxon signed-rank tests follow as post-hoc comparisons; running all pairwise tests without an omnibus would inflate the error rate. Effect sizes are reported as rank-biserial correlations, since a p -value conveys nothing about magnitude. Benjamini–Hochberg correction[12] is applied across the full family of tests rather than within subgroups. Downstream AUC differences are tested with DeLong’s method[13] for correlated ROC curves alongside stratified bootstrap intervals.

3.6 Downstream Utility Protocol

RQ4 is evaluated by training classifiers on 4 training sets differing only in what synthetic data is added: **none** (real data only), **all** (every generated reconstruction), **filtered**

(reconstructions whose mean uncertainty falls below a percentile threshold), and **random_matched** (the same *number* of reconstructions as the filtered arm, drawn at random).

The fourth arm carries the argument. Comparing **filtered** against **none** confounds the value of adding synthetic data at all with the value of choosing which to add. Only the comparison against **random_matched** isolates the second, because the arms differ in selection rule and nothing else, including size. A study reporting only the first comparison cannot distinguish a good selection criterion from a criterion that does nothing while more data does the work.

The filtering threshold is itself a free parameter, so the comparison is repeated across percentiles spanning aggressive to permissive filtering. At every threshold the control is size-matched, which differences out the dataset-size effect point by point; what the resulting curve responds to is the selection criterion alone.

Architecture, optimizer, schedule and augmentation policy are shared across all arms, and positive classes are weighted to compensate for prevalence imbalance. Decision thresholds are selected on the validation split, then applied unchanged to the test split.

3.7 Replication Protocol

Downstream comparisons of this kind are typically reported from a single training run with uncertainty quantified by bootstrap resampling of the test set. That captures one source of variability and omits another: two identically configured runs differing only in random seed produce different initialisations, batch orderings, dropout masks and selected checkpoints, and therefore different test performance.

The principal downstream comparison was therefore replicated across five independent training seeds per arm. The analysis rule was fixed before any run was executed: compute the per-seed difference between arms for each class; report the mean and standard deviation across seeds, the latter being the empirical training-noise floor; treat an effect as surviving only if its sign is consistent across all seeds and its magnitude is large relative to that standard deviation; and treat a sign change across seeds as evidence that the single-run estimate reflected training noise, irrespective of the mean. Specifying the rule in advance is what makes the outcome interpretable, since otherwise a failed replication can be reframed after the fact as an inconclusive but suggestive mean.

Only the training seed is varied. The data split, augmentation selection and all hyperparameters are held fixed, so the measured variability is attributable to training stochasticity alone. Two sources therefore remain unmeasured: selection noise, since the augmentation draw is fixed, and generation-seed variability, since all maps come from a single generation run. Both are recorded in Section 7.5.

3.8 Reproducibility

Pretrained classifier weights are pinned to a fixed local copy rather than downloaded at run time, so replication arms share byte-identical initialisation with the runs they replicate. Splits are generated from a fixed seed and reused across every stage. Maps, checkpoints, per-image metric tables and statistical outputs are retained as versioned artefacts; code is released as described in Appendix B.

Chapter 4

Experimental Setup

4.1 Dataset

All experiments use VinDr-CXR, a chest radiograph dataset with radiologist-drawn bounding-box annotations for local findings and a predefined train/test partition. It is used because it provides spatial annotations rather than image-level labels alone, without which localisation cannot be evaluated at all. Training-partition annotations come from multiple independent readers. No adjudication is applied; the union of all readers' boxes is taken as the region of interest, retaining findings on which readers disagreed. Images are converted from DICOM, windowed, resized and stored in a single HDF5 archive. Box coordinates are retained at original resolution and rescaled at evaluation time, so no resampling error enters the stored annotations.

4.1.1 Class Selection

VinDr-CXR annotates a large number of distinct local findings. Five are used here: Pneumothorax, Consolidation, Nodule/Mass, Cardiomegaly and Atelectasis. The subset spans the properties expected to affect localisation. Cardiomegaly is large, central, and defined by the size of an anatomical structure and not by a texture abnormality. Nodule/Mass is small and focal. Consolidation and Atelectasis are intermediate in extent with characteristic textural appearances. Pneumothorax is peripheral and defined by absent rather than present markings. This spread is what makes cross-class comparison informative.

The restriction to five is a compute constraint, not a methodological one: each class requires its own generation run, and the available budget went to replication rather than breadth. Extending to the full annotated set is the first item of future work.

4.1.2 Splits and Leakage Control

Images are partitioned from a fixed seed and the partition reused unchanged at every stage. Validation is used only for threshold and checkpoint selection; the test split only

for final evaluation.

Contamination is checked in both directions, as an assertion rather than a report, so a violation halts the pipeline. In the downstream experiments, no image contributing to a generated augmentation set may appear in the test split. The held-out map evaluation reverses this: every evaluated map must correspond to a test-split image, and none may appear among the images used to generate training augmentation. A check written for one direction passes silently in the other, which is why both are implemented.

Table 4.1: Dataset composition by class. Test positives are images carrying at least one annotated box for the class, and are the sample sizes for all per-class statistics.

Class	Train	Test positives	Test prevalence
Pneumothorax	65	18	2.7%
Atelectasis	138	27	4.1%
Consolidation	254	54	8.2%
Nodule/Mass	578	117	17.7%
Cardiomegaly	1622	331	50.2%
Total images	2657	660	

The prevalence range spans roughly 2.7% to 50%, with two consequences carried through the results. Threshold-dependent metrics are not comparable across classes without accounting for prevalence, which is why AUPRC is always reported against the base rate. And statistical power differs by an order of magnitude: an effect detectable on Cardiomegaly’s 331 positives may be invisible on Pneumothorax’s 18.

4.2 Models and Configuration

The generative model is a latent diffusion model fine-tuned for chest radiograph synthesis, used zero-shot without further fine-tuning, RoentGenV2 model[14]. Using it zero-shot keeps the estimator free of exposure to the labels or images used for evaluation, which is what makes the comparison in Section 5.4 interpretable. The classifier used for both the comparator maps and the downstream experiments is DenseNet-121,[15] initialised from ImageNet-pretrained weights with a multi-label head over the five target classes.[15] Implementation uses PyTorch[16] with the Hugging Face `diffusers` library [17]; analysis uses `scipy`[18], `scikit-learn`[19] and `statsmodels`[20].

4.3 Compute Budget

Experiments ran on a session-limited dual-GPU environment with 16GB vram and storage limitations, which shaped several decisions recorded here because they are otherwise hard to justify. Each classifier training run takes approximately thirty minutes, so the five-seed replication of two arms costs ten runs and each additional filtering

Table 4.2: Experimental configuration. Values marked *varied* are the ablation axes of Section 3.3.1; all others are constant across every experiment reported.

Parameter	Value	Note
Resolution	512×512	Maps computed at this resolution
Ensemble size N	20 (<i>varied</i>)	Reduced to 10 as negative control
Stochasticity η	1.0 (<i>varied</i>)	Ablated over $\{0.0, 0.5, 1.0\}$
Denoising steps	30(<i>varied</i>)	Reduced setting of 15 in ablation
Guidance scale	1.0	Held fixed; not ablated
Inversion	DDIM, $\eta = 0$	Latent cached per image
Architecture	DenseNet-121	ImageNet-initialised
Optimiser / LR	Adam / $1e^{-10}$	
Batch / epochs	8/15	Checkpoint selected on validation
Dropout	0.3	Also used for MC dropout passes
Loss	Weighted BCE	Positive weighting, capped
Data seed	42 (fixed)	Splits and selection constant
Training seeds	42, 7, 123, 2024, 31337	Replication protocol
Bootstrap	2000 resamples	Resampled over images
Significance	0.05	Benjamini–Hochberg corrected
Percentiles	25, 50, 75, 90	Threshold dose–response

percentile costs two more.

The same budget explains the two most visible restrictions of this study: the class set was limited to five, and uncertainty maps were generated once per image rather than across multiple generation seeds. The allocation favoured replicating the downstream comparison over extending its breadth, on the grounds that an unreplicated result across more classes would compound the problem identified in Section 3.7 rather than reduce it.

Chapter 5

Localisation of Generative Uncertainty

This chapter reports whether pixel-wise variance across inversion-anchored diffusion ensembles agrees spatially with expert annotations (RQ1), how that agreement responds to the sampling configuration (RQ2), and how the estimator compares with classifier-derived uncertainty and supervised saliency (RQ3). All figures are per-image estimates aggregated across images, with intervals from bootstrap resampling over images as described in Section 3.5.3.

5.1 Qualitative Characterisation

Before quantitative evaluation it is worth establishing what the maps look like. Figure 5.1 shows representative variance maps alongside their source radiographs and annotation overlays.

Two features are consistent across classes. Variance is highest along high-contrast anatomical boundaries, particularly the rib margins, the diaphragm and the mediastinal border. This is expected: small differences in reconstructed edge position produce large per-pixel differences, so a sharp boundary generates variance whether or not the model is genuinely uncertain about the anatomy there. Variance is lowest in homogeneous regions — the soft tissue of the shoulders, the unstructured portions of the lung fields.

The variance signal therefore contains a substantial edge-driven component unrelated to pathology. Any localisation performance the maps achieve is achieved in spite of it, and what the quantitative evaluation measures is the degree to which pathology-related variance rises above it.

The consequence is that the variance signal contains a substantial edge-driven component that is unrelated to pathology. Any localisation performance the maps achieve is achieved in spite of this component, and the degree to which pathology-related vari-

Representative pixel-wise variance maps

Variance concentrates along anatomical edges (diaphragm, rib margins) regardless of pathology class -- the visual basis of the dose--response and pointing-hit results in Section 5.1--5.2. Multiple boxes on a row reflect independent radiologist annotations of the same finding, not a rendering artefact.

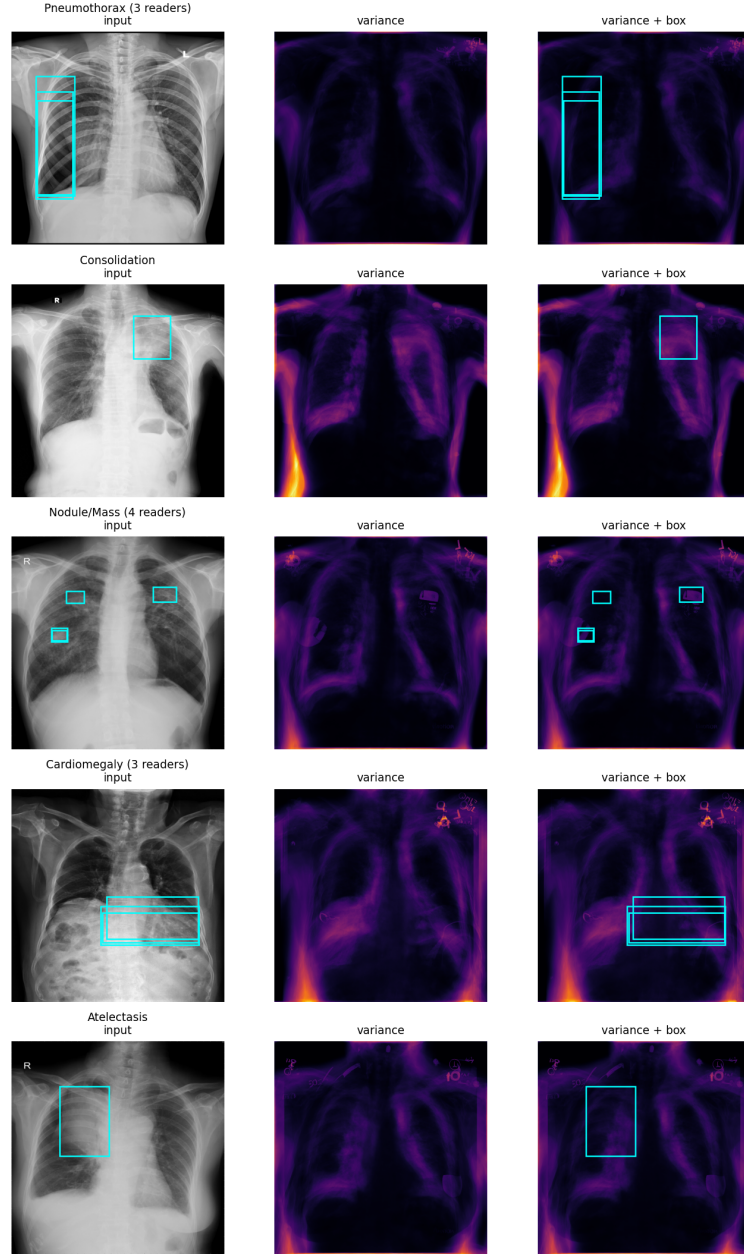


Figure 5.1: Representative pixel-wise variance maps. Each row is a pathology class; columns show the input radiograph, the variance map computed across the reconstruction ensemble, and the map with radiologist bounding boxes overlaid. Variance concentrates along high-contrast anatomical boundaries in every case, and the degree to which it additionally concentrates within annotated regions varies by class.

ance rises above it is what the quantitative evaluation measures.

5.2 Spatial Agreement with Expert Annotations

Table 5.1 and Figure 5.2 report per-image pixel AUROC by class on the held-out test split.

Table 5.1: Per-image pixel AUROC by pathology class on the held-out test split, with bootstrap confidence intervals over images. n is the number of test images carrying at least one annotated box for the class. AUPRC is reported alongside the mean in-box pixel fraction, since box coverage rather than 0.5 is the relevant baseline. Frac. above chance is the proportion of individual images on which the map ranked in-box pixels above out-of-box pixels; a value well below 0.5 indicates the map is systematically anti-correlated with the annotation, not merely uninformative.

Class	n	AUROC	95% CI	AUPRC	Frac. above chance
Consolidation	54	0.694	[0.664, 0.723]	0.065	0.926
Atelectasis	27	0.635	[0.599, 0.672]	0.083	0.926
Nodule/Mass	117	0.620	[0.590, 0.648]	0.027	0.795
Pneumothorax	18	0.573	[0.508, 0.630]	0.106	0.722
Cardiomegaly	331	0.450	[0.442, 0.459]	0.061	0.266

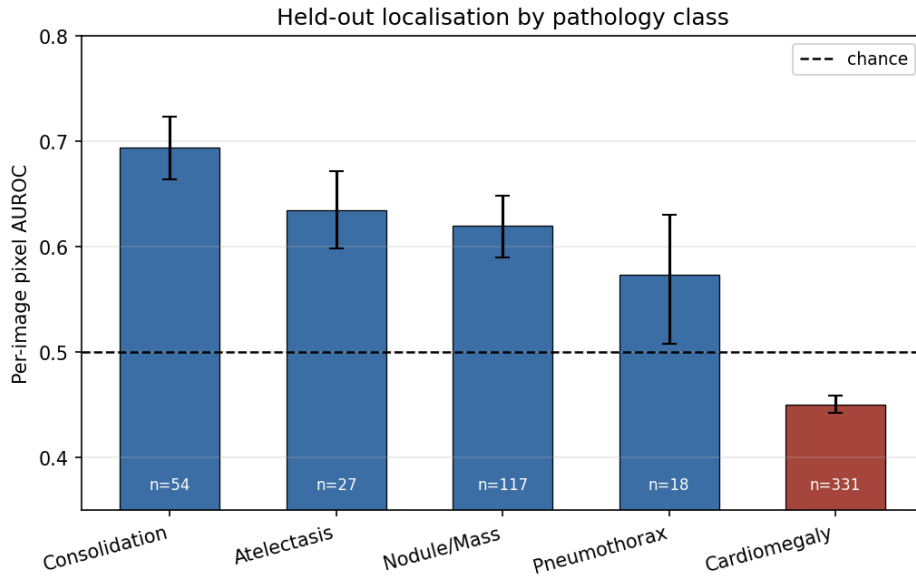


Figure 5.2: Per-image pixel AUROC by pathology class on the held-out test split. Error bars are bootstrap confidence intervals over images; the dashed line marks chance. Cardiomegaly’s interval sits entirely below chance rather than merely overlapping it.

RQ1’s answer splits the five classes into two groups, not one graded scale. Consolidation, Atelectasis, Nodule/Mass and Pneumothorax localise above chance, with confidence intervals excluding 0.5 across the board. Cardiomegaly does not merely fail to localise: its interval, [0.442, 0.459] on 331 images, sits entirely *below* chance. The fraction-above-chance figures explain why. For the four classes that do localise, individual images rank in-box pixels above out-of-box ones 72–93% of the time. For

Cardiomegaly it is 27% – the map ranks the annotation the wrong way round on roughly three images in four.

That is a stronger, more specific claim than the near-chance result from the training-split evaluation, and it sharpens the account in Section 7.2. An uninformative signal would be consistent with the model simply having nothing to say about a class. An anti-correlated one says more: whatever the variance is responding to is concentrated *outside* the annotated region. The qualitative pattern from Section 5.1 explains why – variance peaks along rib and diaphragm edges rather than within smooth tissue, and Cardiomegaly’s box encloses a large, texturally uniform cardiac silhouette. The boundary structures driving the model’s instability fall mostly outside it.

A second statistic makes the same point more bluntly. Across all 547 test images spanning the five classes, the single highest-variance pixel fell inside an annotated box in **zero** of them. Not a low proportion – none. Whatever the ranking captures near the top of the distribution, it is not the location of peak instability. That is consistent with peak variance being an edge phenomenon on every image, not a lesion-adjacent one on most.

5.3 Sensitivity to Sampling Configuration

RQ2 asks how the signal depends on the configuration that produces it. Figure 5.3 and Table 5.2 report variance magnitude and localisation performance across the three ablation axes described in Section 3.3.1.

Table 5.2: Sampling configuration ablation. Change in variance magnitude and in localisation AUROC relative to the reference configuration ($\eta = 1.0$, full step count, $N = 20$). Reducing the ensemble size is a negative control: it changes only the number of samples the variance is estimated from, not the distribution being sampled.

Axis	Setting	Δ variance	Δ AUROC	Role
Stochasticity	$\eta = 0.5$	−47%	+0.037	Primary
Steps	15 steps	−32.4%	+0.016	Primary
Ensemble size	$N = 10$		≈ 0	Negative control

The pattern is consistent across both primary axes and is the most informative result in this chapter. Halving the stochasticity parameter removes roughly 47% of the variance signal and improves localisation AUROC by 0.037. Reducing the step count removes roughly a third of the variance and improves AUROC by 0.016. The relationship is monotonic in the settings tested: the configuration that suppresses more variance yields the larger improvement.

The negative control behaves as designed. Halving the ensemble size returns a null result on both axes, which confirms that the pattern observed on the primary axes is a property of the sampling process rather than an artefact of the estimator’s sample size.

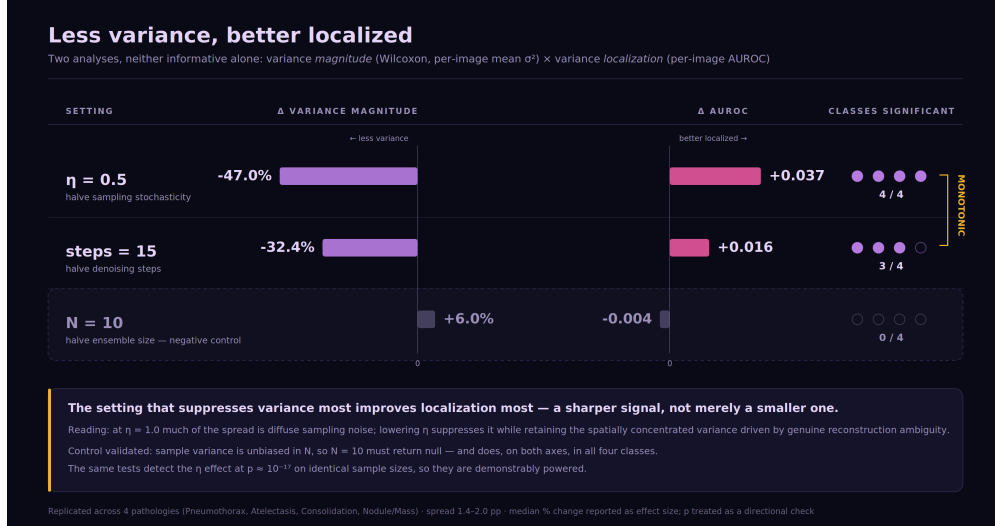


Figure 5.3: Dose–response between variance magnitude and localisation performance. Configurations that substantially reduce the magnitude of the variance signal do not reduce its spatial agreement with annotations, and in the settings tested slightly improve it. The negative-control condition, which reduces only the number of samples the estimate is drawn from, produces no change on either axis.

Neither primary axis produces this result in isolation in a way that would be convincing. A single configuration showing improved AUROC at reduced variance could be a chance fluctuation. Two independent axes producing the same directional effect, with a third axis designed to produce nothing producing nothing, is a coherent dose–response.

The interpretation is that a substantial fraction of the raw variance signal does not carry localisation information. Removing it improves the ratio of signal to background, which is consistent with the qualitative observation in Section 5.1 that much of the variance is edge-driven. This finding is developed in Section 7.1.

5.4 Comparison with Classifier-Based Uncertainty

RQ3 compares the generative estimator against Monte Carlo dropout logit variance and against Grad-CAM attribution variance under dropout, all evaluated on the same held-out images.

The omnibus test rejects a shared ordering across methods in four of five classes – Pneumothorax is the exception, likely just a sample-size effect at $n = 18$. The post-hoc comparisons are consistent: **MCD-logit variance beats diffusion variance in every class**, reaching significance after correction in four of five. Effect sizes range from moderate (Atelectasis, rank-biserial -0.56) to near-total separation on Cardiomegaly, where every single held-out image favoured MCD-logit (-1.00).

This result forces a revision of Section 3.4.1. The held-out comparison was designed to test whether Stage 4’s finding – MCD-logit beating diffusion “nearly everywhere” – was an artefact of the classifier having trained on the images it was later evaluated on. Mostly, it was not. Moving to images neither model has seen leaves MCD-logit’s

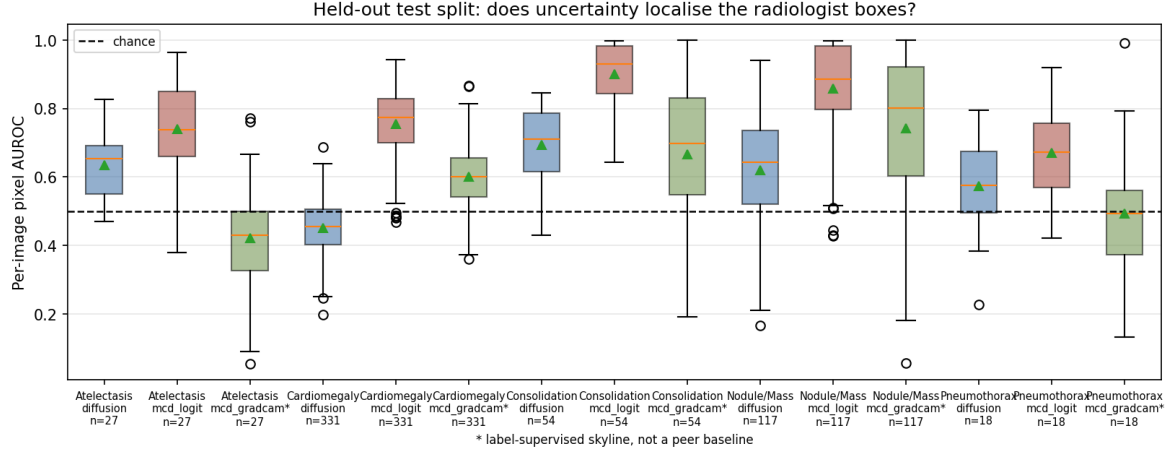


Figure 5.4: Distributions of per-image pixel AUROC by method and pathology class on the held-out test split. All three map types were evaluated at native 512×512 resolution, so no resampling confound applies to this comparison. Boxes show the interquartile range across images; the dashed line marks chance.

Table 5.3: Friedman omnibus test across the three methods, on held-out images all three cover. Mean rank is computed per image (1 = best of the three); a value near 2 indicates no consistent ordering. p -values are Benjamini–Hochberg corrected across the full pair $\times$ class $\times$ metric family reported in this chapter.

Class	n	Mean rank <i>diffusion</i>	Mean rank <i>mcd_logit</i>	Mean rank <i>mcd_gradcam</i>	p_{BH}
Pneumothorax	18	1.833	1.722	2.444	0.082
Atelectasis	27	1.889	1.296	2.815	2.3×10^{-7}
Consolidation	54	2.426	1.148	2.426	4.3×10^{-13}
Nodule/Mass	117	2.581	1.410	2.009	1.3×10^{-17}
Cardiomegaly	331	2.952	1.012	2.036	4.6×10^{-135}

Table 5.4: Post-hoc paired comparisons on pixel AUROC, Benjamini–Hochberg corrected across the full family. Rank-biserial correlation is the effect size; negative values favour the second-named method.

Class	Mean Δ	95% CI	Rank-biserial	p_{BH}
<i>diffusion</i> – <i>mcd_logit</i>				
Pneumothorax	−0.097	[−0.191, −0.004]	−0.222	0.118
Atelectasis	−0.105	[−0.175, −0.030]	−0.556	0.016
Consolidation	−0.206	[−0.242, −0.170]	−0.926	1.9×10^{-9}
Nodule/Mass	−0.238	[−0.276, −0.197]	−0.744	1.4×10^{-15}
Cardiomegaly	−0.305	[−0.314, −0.296]	−1.000	1.5×10^{-54}
<i>diffusion</i> – <i>mcd_gradcam</i>				
Pneumothorax	+0.081	[−0.012, 0.165]	+0.556	0.087
Consolidation	+0.028	[−0.031, 0.090]	+0.074	0.441
Nodule/Mass	−0.122	[−0.177, −0.069]	−0.419	1.0×10^{-4}
Cardiomegaly	−0.150	[−0.160, −0.140]	−0.903	1.2×10^{-52}
Atelectasis	+0.214	[0.143, 0.291]	+0.778	1.2×10^{-5}

advantage intact, and on the classes with the most test images, overwhelming. The image-level confound from Stage 4 was real and worth removing. Removing it barely moved the gap. What is left is the asymmetry already flagged as a limitation: MCD-logit comes from a network trained on the very label set the annotations encode, while the generative estimator has seen neither these images nor any pathology label. Label supervision, not image exposure, looks like the real driver – and this study’s design cannot correct for it.

Grad-CAM variance under dropout does not behave like a clean upper bound on what a supervised method can achieve – which is itself informative. It beats diffusion on Nodule/Mass and Cardiomegaly, is indistinguishable from it on Pneumothorax and Consolidation, and loses to it decisively on Atelectasis (rank-biserial $+0.78$, favouring diffusion). It loses to MCD-logit in every class. That unevenness fits what the quantity actually is: the variance of Grad-CAM attribution across repeated dropout passes, not a single deterministic saliency map (Section 3.4). Attribution dispersion under perturbation is a different question from attribution itself – a region can attract inconsistent evidence under dropout without being the lesion. Its patchy performance here suggests it is simply a noisier quantity than classical Grad-CAM, not a privileged reference. Read “skyline” as naming this comparator’s label-supervised origin, then, not as a claim that it upper-bounds performance. On this evidence it does not.

The native-resolution audit in Table A.1 rules out one alternative explanation. All three map types were generated or stored at 512×512 , so none of the comparisons above trace back to a coarse map being upsampled into an artificially smooth, high-overlap region. The gap between diffusion and MCD-logit is a property of the estimators, not the preprocessing.

Chapter 6

Downstream Utility and Replication

This chapter tests whether the uncertainty signal has practical value as a data curation criterion (RQ4), and then reports a pre-registered replication of that test. The replication is presented as a result in its own right rather than as an appendix to the first analysis, because its outcome required withdrawing a finding and because its implications extend beyond this study.

6.1 Classification Performance by Arm

Four classifiers were trained on identical architectures and schedules, differing only in the synthetic data added to the real training set, as defined in Section ?? . Table 6.1 reports their performance on the held-out test split.

Table 6.1: Downstream classification performance by augmentation arm on the held-out test split, from a single training run. AUPRC lift is AUPRC divided by class prevalence, which is the relevant baseline given the prevalence range in Table 4.1. Figures in this table are superseded by the replication reported in Section 6.4 and should not be read in isolation.

Class	Arm	Prevalence	AUROC	AUPRC	AUPRC lift
Pneumothorax	none	2.7%	0.865	0.560	20.5
	all		0.884	0.400	14.7
Atelectasis	none	4.1%	0.924	0.466	11.4
	all		0.840	0.251	6.1
Consolidation	none	8.2%	0.874	0.426	5.2
	all		0.873	0.362	4.4
Nodule/Mass	none	17.7%	0.716	0.352	2.0
	all		0.685	0.320	1.8
Cardiomegaly	none	50.2%	0.935	0.936	1.9
	all		0.938	0.934	1.9

The AUPRC lift column is included because raw AUPRC is not interpretable across a prevalence range spanning 2.7% to 50.2%. A Pneumothorax AUPRC of 0.560 repre-

sents a twenty-fold improvement over the base rate, while a Cardiomegaly AUPRC of 0.936 represents less than a doubling. Reporting AUPRC without the corresponding base rate would invert the apparent ranking.

6.2 Filtering Versus Size-Matched Random Selection

The comparison that answers RQ4 is between the **filtered** arm and the **random_matched** arm, which differ in selection rule and in nothing else including size.

The result is null in all five classes. Selecting synthetic images by low generative variance produced no measurable advantage over selecting the same number of images at random. No class showed a difference whose confidence interval excluded zero, and no class showed a difference that survived correction for multiple comparisons.

This is a substantive answer to RQ4 rather than a failed experiment, and the size-matched design is what makes it so. Had the comparison been against the unaugmented arm, any difference would have confounded the value of the selection criterion with the value of adding data. Against a size-matched control, the selection criterion is isolated, and it does nothing.

6.3 Filtering Threshold Dose-Response

A single threshold cannot distinguish a criterion that is inert from one whose benefit depends on how aggressively it is applied. The comparison was therefore repeated across filtering percentiles from aggressive to permissive.

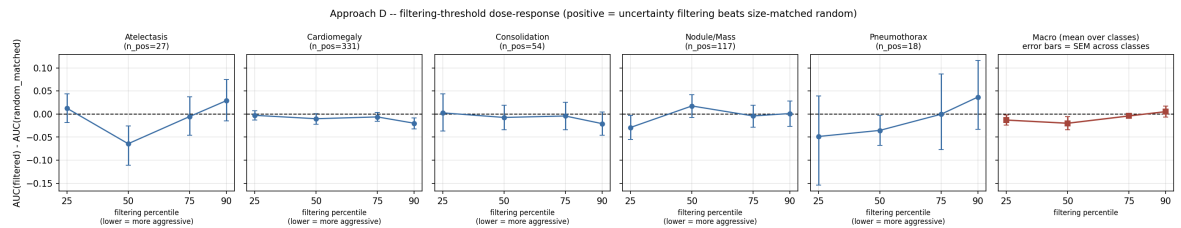


Figure 6.1: Difference in test AUC between uncertainty-filtered and size-matched random augmentation, across filtering thresholds. Positive values favour uncertainty-based selection. Because the control is size-matched at every threshold, the dataset-size effect is differenced out at each point and the curve responds to the selection criterion alone.

The result is flat everywhere. At every threshold tested and in every class, the bootstrap confidence interval on the filtered-minus-random- matched difference includes zero, and the macro panel in Figure 6.1 moves within roughly 0.02 AUC of zero with no consistent direction as filtering goes from aggressive to permissive.

One nominal exception needs addressing head-on, given the argument this thesis makes elsewhere about single-run statistics. The regression of the difference on percentile for Pneumothorax returns $p = 0.040$ with $R^2 = 0.921$ – on the surface, a clean

Table 6.2: Filtered-minus-random-matched AUC difference by class and filtering percentile.

Class	p25	p50	p75	p90
Atelectasis	+0.013	−0.064	−0.005	+0.029
Cardiomegaly	−0.003	−0.010	−0.006	−0.020
Consolidation	+0.003	−0.007	−0.004	−0.021
Nodule/Mass	−0.029	+0.018	−0.004	+0.001
Pneumothorax	−0.049	−0.035	−0.000	+0.037

Table 6.3: Per-class linear trend of the difference against percentile. Four points per class, so treat the fit descriptively – p_{BH} is Benjamini–Hochberg corrected across the five class-level tests.

Class	Slope/10pt	R^2	p	p_{BH}	Same sign
Pneumothorax	+0.013	0.921	0.040	0.200	No
Consolidation	−0.003	0.700	0.164	0.375	No
Cardiomegaly	−0.002	0.600	0.225	0.375	Yes
Nodule/Mass	+0.003	0.242	0.508	0.635	No
Atelectasis	+0.004	0.079	0.718	0.718	No

trend across the four points tested. It is not a trend worth reporting. The regression has only two degrees of freedom at four points. Its sign is not even consistent across thresholds (Table 6.3, `all_same_sign` = No). And running an independent slope test in five classes at $\alpha = 0.05$ is exactly the setup where chance alone predicts one nominal hit. Correcting across the five class-level tests with Benjamini–Hochberg raises Pneumothorax’s p -value to 0.200. Treating the uncorrected number as a finding would repeat, in miniature, the mistake the seed replication in Section 6.4 exists to catch.

Two cautions apply to reading this curve. Each point derives from a single training run, so the intervals shown cover test-set sampling variability and not the training variability quantified in Section 6.4. And because the percentile cuts are nested, adjacent thresholds share most of their selected images, so the points are not independent observations.

The threshold sweep does not qualify the null result in Section 6.2 – it reinforces it. Uncertainty-based filtering does nothing distinguishable from a size-matched random draw, at any aggressiveness tested, from the most permissive threshold to the most aggressive.

6.4 Seed Replication

6.4.1 Motivation and Pre-registered Rule

The single-run comparison in Table 6.1 showed the augmented arm underperforming the unaugmented arm on Atelectasis by 0.084 AUROC, with a bootstrap confidence interval excluding zero and a p -value surviving correction within its family. Taken at

face value this was the strongest downstream finding in the study, and it would have been reported as evidence that unfiltered synthetic augmentation actively degrades performance on some classes.

It was obtained from one training run. The replication described in Section 3.7 was executed to test whether an effect of that magnitude is separable from training noise, with the decision rule fixed in advance of the runs.

6.4.2 Seed-to-Seed Variability

Table 6.4 gives the per-seed differences; Figures 6.2 and 6.3 display them two ways – every seed plotted individually against the original estimate, then collapsed to a mean with its interval.

Table 6.4: Change in test AUROC from adding unfiltered synthetic augmentation, by class and training seed. Data, augmentation set, splits and every hyperparameter are identical across columns; only the training seed differs. The single-run estimate originally reported is the seed 42 column.

Class	Seed 7	Seed 42	Seed 123	Seed 2024	Seed 31337
Atelectasis	−0.001	− 0.084	−0.008	−0.019	+0.012
Cardiomegaly	+0.003	+0.003	−0.004	−0.013	−0.004
Consolidation	−0.011	−0.001	−0.001	−0.003	+0.009
Nodule/Mass	+0.011	−0.031	−0.001	−0.043	−0.035
Pneumothorax	+0.032	+0.019	+0.065	−0.008	+0.035

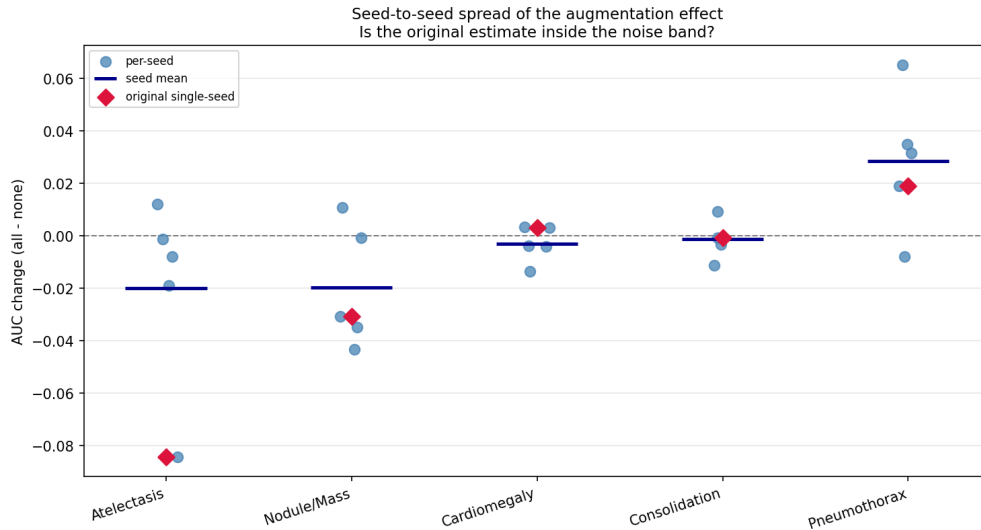


Figure 6.2: Change in test AUROC from adding unfiltered synthetic augmentation, one point per training seed, by class. The navy bar marks the cross-seed mean; the red diamond marks the seed-42 estimate originally reported. On Atelectasis the original estimate sits visibly outside the cluster of the other four seeds; on every other class it sits inside it.

The pre-registered rule returns an unambiguous verdict. Every class changes sign across seeds. No class has a mean whose confidence interval excludes zero, before any correction for the five comparisons (Figure 6.3). The -0.084 estimate falls outside the

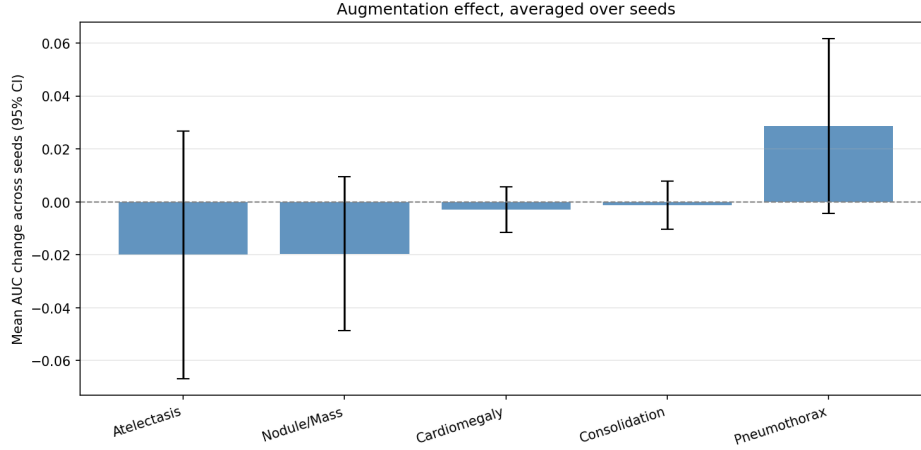


Figure 6.3: Mean change in test AUROC from unfiltered augmentation, averaged across five training seeds, with 95% confidence intervals. Every interval crosses zero.

Table 6.5: Cross-seed summary. The standard deviation across seeds is the empirical training-noise floor for each class, and is the quantity against which the single-run estimate must be judged. Confidence intervals are t -based with four degrees of freedom.

Class	Mean	Seed SD	95% CI	Single run	Sign stable?
Atelectasis	-0.020	0.038	$[-0.067, +0.027]$	-0.084	No
Nodule/Mass	-0.020	0.023	$[-0.049, +0.010]$	-0.031	No
Pneumothorax	+0.029	0.027	$[-0.004, +0.062]$	+0.019	No
Consolidation	-0.001	0.007	$[-0.010, +0.008]$	-0.001	No
Cardiomegaly	-0.003	0.007	$[-0.012, +0.006]$	+0.003	No

interval of the very quantity it was estimating.

6.4.3 Diagnosis of the Original Estimate

The failure is more specific than sampling variability in general. Seed 42 did not produce an unusually poor augmented run. It produced an unusually good unaugmented one: its Atelectasis AUROC in the unaugmented arm was 0.924 against a mean of 0.865 across the remaining four seeds, placing it 2.86 standard deviations above them. Its augmented run, at 0.840, sat near the bottom of that arm’s range. The reported difference was the product of two independent extremes coinciding in one run.

A second observation follows from the same data. The unaugmented arm is roughly twice as variable across seeds as the augmented arm on this class (standard deviations of 0.034 and 0.018 respectively). Adding synthetic data did not improve mean performance, but it appears to have stabilised training. That is a more defensible observation than the one it replaced. It rests on five runs and one class, though, and is offered as an observation rather than a claim.

6.4.4 Implications for Single-Run Reporting

The comparison between the two uncertainty sources is the transferable result. On Atelectasis, the bootstrap interval from the single run implied a test-sampling standard deviation of approximately 0.028. The measured seed-to-seed standard deviation is 0.038. Training variability exceeds test-set sampling variability on this task, and the two combine to approximately 0.047, nearly double what the bootstrap alone reported.

The bootstrap interval was not merely narrow. It was measuring the wrong thing, or rather only one of the two things that needed measuring. A procedure that resamples the test set while holding the trained model fixed cannot detect variability that arises from training the model differently. In this setting, that is the larger component.

Across the twenty-five class-by-seed bootstrap tests conducted during the replication, five returned intervals excluding zero. That is close to the proportion expected from uncorrected testing on data containing no effect. It illustrates the practical consequence: a study reporting a single run of this comparison would have had a reasonable chance of a significant result in some class, regardless of whether any effect existed.

The finding originally reported is therefore withdrawn. The corrected statement is that unfiltered synthetic augmentation produced no change in test AUROC distinguishable from training noise on any of the five classes, with the largest non-significant trends being a small decrement on Nodule/Mass and a small increment on Pneumothorax.

Chapter 7

Discussion

7.1 Interpreting Generative Uncertainty as a Spatial Signal

The localisation results support a narrower claim than a uniform account across classes would suggest. Variance across inversion-anchored diffusion ensembles agrees with expert annotations above chance for four of five pathology classes; for the fifth, Cardiomegaly, it is systematically anti-correlated with them rather than merely uninformative. Across the full held-out set, the single highest-variance pixel never fell inside an annotated box, in any of 547 images. This is not the signature of a competitive localisation method with a weak spot – it is the signature of a signal dominated by something other than pathology.

The qualitative and dose-response results in Sections 5.1 and 5.3 identify what that something is. Variance concentrates along anatomical edges – principally rib margins, the diaphragm, and the mediastinal border – and configurations that substantially reduce total variance leave localisation agreement intact or improved. The pointing-hit result is the sharpest confirmation available: if peak variance were ever lesion-driven, it would occasionally land inside a box. It does not, on any image tested. Peak instability in these ensembles is an edge phenomenon, not a pathology one. Cardiomegaly’s anti-signal follows directly: its annotation encloses tissue that is *edge-poor* by construction, a large smooth cardiac silhouette whose defining boundary – the cardiothoracic ratio – is a relationship between distant structures, not a local texture.

This points to a practical direction the present study did not pursue. If the signal is dominated by boundary-position instability, decomposing or regressing out the edge component should raise the ratio of pathology-related to background variance more efficiently than tuning the sampler further. It should also help Cardiomegaly least of the five classes – its finding was never encoded in local structure to begin with.

The comparison against classifier-derived uncertainty in Section 5.4 bears on how

this signal should be read against alternatives. MCD-logit variance outperforms the generative estimator in every class tested, with effect sizes that in several cases approach complete separation – and this held even after the image-level confound identified in Stage 4 was removed. The residual explanation, developed in Section 3.4.1, is label supervision rather than image exposure, and it is arguably the more important finding of the two. Most of Stage 4’s original gap was never about data leakage. An unsupervised generative signal, on this evidence, is not yet a substitute for a supervised one when boxes are available to train on.

7.2 Why Localisation Varies Across Pathologies

The spread across classes was wider than the uncertainty on any single class, which makes the pattern the finding. Cardiomegaly’s position at the bottom of the range is the most instructive case, because the obvious explanation turned out to be wrong.

The initial hypothesis was textural: that the generative model reconstructs the cardiac region with unusually little variation, suppressing the signal where the annotation lies. Direct testing rejected this. The class retains as much reconstructed texture as any other, and the suppression the hypothesis required is not present.

The account that survives concerns the nature of the annotation. Cardiomegaly is not a local appearance abnormality. It is a judgement that a normal structure is too large relative to the thorax, and the box drawn around it encloses tissue that looks unremarkable. A method that responds to local appearance instability has nothing distinctive to respond to inside that box, because there is no local abnormality there. The finding is real and diagnostically important, but it is encoded in a global geometric relationship rather than in the pixels the box contains.

If this account is correct it predicts a general pattern: appearance-defined findings should localise better than size- or position-defined ones. Within appearance-defined findings, smaller and more focal lesions should localise better than diffuse ones. The five classes studied here are consistent with that ordering but cannot establish it. Five points cannot support a regression of localisation performance on lesion geometry. This is the strongest argument for extending the class set, and it is why breadth appears first in Section 8.3.

7.3 Uncertainty Filtering and Downstream Utility

The downstream result is null and, given the design, informative. Selecting synthetic images by low generative variance produced no advantage over selecting the same number at random, in any class, at any filtering threshold tested.

Two claims must be kept apart here, and the size-matched control is the reason they can be. The first is that adding synthetic data helps or harms. The second is

that choosing which synthetic data to add helps. This study finds no support for the second. It says less about the first: the replication in Section 6.4 shows that the effects at issue there are smaller than training noise in this setting. The honest summary is that neither claim is supported by the present evidence.

Several explanations are compatible with the null. Mean variance over a reconstruction may simply be a poor proxy for what makes a synthetic image useful for training – the criterion could be wrong rather than the idea. It’s possible the synthetic images are similar enough in quality that no selection rule could usefully separate them. Or augmentation volume may simply be too small, relative to the real training set, for any selection effect to register. The present design cannot distinguish between these, and doing so would mean varying augmentation volume as its own axis.

What the result establishes is a methodological point about how such claims should be tested. A study comparing filtered augmentation against no augmentation would, on this data, have reported differences of the kind shown in Table 6.1 and attributed them to the selection criterion. The size-matched control is inexpensive, and it changes the conclusion.

7.4 Methodological Implications

The replication result generalises beyond this study, and it is the contribution most likely to be useful to others.

Reporting a comparison from a single training run with a bootstrap confidence interval is standard practice.[21] The procedure is internally valid but incomplete: it quantifies variability due to the finite test set while holding the trained model fixed, and therefore cannot see variability due to training the model differently. This study measured both components on the same task and found the unmeasured one to be the larger.

That does not make bootstrap intervals wrong – it makes them intervals on a different quantity than the one readers take them to describe. Read a bootstrap interval on a single-run difference as an interval on the effect of the intervention, and you are making an inference the procedure does not support. In this study, that inference would have produced a published false positive.

Three recommendations follow. Comparisons of this kind should report a seed-level standard deviation alongside any test-set interval. Where compute does not permit full replication of every comparison, replicating one representative comparison establishes a noise floor the rest can be read against. That is inexpensive relative to the cost of a retracted claim. Decision rules should be fixed before replication runs are executed – otherwise a failed replication becomes an inconclusive but suggestive mean.

7.5 Limitations

The following limitations bound every claim in this thesis.

Single dataset. All results come from one dataset, from one population and one set of acquisition protocols. Nothing here establishes transfer to other institutions or scanner distributions.

Five classes. Five of the dataset’s annotated local findings were studied. The cross-class pattern in Section 7.2 is suggestive at this scale and cannot be tested as a hypothesis about lesion geometry.

Single generation run. Every uncertainty map derives from one ensemble generated with one sampling seed. The variability of the maps themselves across generation seeds is unmeasured. Training-seed variability turned out larger than expected in the downstream analysis, so assuming generation-seed variability is negligible would be inconsistent.

Fixed guidance. Guidance scale was held at unity throughout and not ablated, so one axis of the sampling configuration remains untested.

Selection noise untested. The augmentation selection was held fixed across all replication runs. The variability that would arise from a different random draw of the size-matched control is not measured at any seed count.

Checkpoint selection on a small validation split. Checkpoints were selected on a validation split containing few positives for the rarer classes. That makes the selected epoch close to arbitrary for those classes, and plausibly contributes to the training variability reported in Section 6.4. This mechanism was not tested directly.

Residual label supervision in comparators, confirmed rather than hypothetical. Moving the method comparison to held-out images removed the image-level confound in Stage 4, but not the label-supervision asymmetry argued in Section 3.4.1. This is no longer a stated risk. The held-out results in Section 5.4 show MCD-logit outperforming diffusion variance in every class, with effect sizes reaching complete separation on Cardiomegaly, even with image exposure equalised. The gap between an unsupervised generative signal and a supervised discriminative one is substantial on this evidence, and this study’s design does not close it.

Bounding-box ground truth. Boxes are axis-aligned and enclose normal tissue alongside the finding, so all spatial metrics measure agreement with box annotations rather than with true lesion extent, and no method can reach a perfect score against them.

Statistical power varies by an order of magnitude across classes. Test positive counts range from 18 to 331. Null results on the rarest classes are weak evidence of absence.

Chapter 8

Conclusion and Future Work

8.1 Summary of Findings

We asked whether the sampling behaviour of a diffusion model can be turned into a spatially resolved account of its uncertainty about a real radiograph, and whether that account is worth having.

RQ1. Pixel-wise variance across inversion-anchored ensembles agrees with expert annotations above chance for several pathology classes, but the agreement is modest and varies substantially across classes. The variation is wider than the uncertainty on any single class, which makes the cross-class pattern the substantive result. Cardiomegaly localises worst, and the explanation lies in the nature of the finding rather than in the model’s reconstruction of it.

RQ2. Localisation agreement is robust to substantial reductions in variance magnitude. Two independent configuration axes each reduced variance by a third or more while leaving agreement intact or slightly improved, with a negative control returning null on both measures. A large fraction of the raw variance signal does not carry localisation information.

RQ3. Evaluated on held-out images, the unsupervised generative estimator no longer suffers from the image-level confound that invalidated the earlier comparison. It still loses. MCD-logit variance outperforms diffusion variance in every class tested, with effect sizes reaching complete separation on Cardiomegaly. The residual explanation is label supervision, not image exposure – MCD-logit is trained on the label set the annotations encode; the generative estimator is not. That asymmetry, not the confound this test was designed to remove, is now the dominant explanation for the gap. It is not correctable within this study’s design.

RQ4. Selecting synthetic images by low generative variance conferred no advantage over selecting the same number at random, in any class or at any filtering threshold tested. A pre-registered five-seed replication of the related augmentation comparison showed that training variability exceeds test-set sampling variability on this task, and

required withdrawing a single-run finding that had appeared statistically significant.

8.2 Contributions Revisited

The methodological contributions have proved more durable than the empirical ones. Inversion anchoring makes the uncertainty estimate specific to an input image, not just descriptive of the generator in general. Per-image analysis produces valid intervals where pixel-pooled inference does not. The size-matched control separates the value of a selection criterion from the value of additional data, and changes the conclusion of the downstream analysis. The replication protocol establishes an empirical noise floor – and this study’s own results show exactly what happens when one is missing.

The empirical findings are appropriately modest: a real but limited spatial signal, a dose–response indicating that much of it is not informative, and a null result on downstream curation obtained under a design that makes the null interpretable.

8.3 Future Work

Broader pathology coverage. Extending generation and evaluation to the full set of annotated local findings would convert the cross-class observation in Section 7.2 from a pattern across five points into a testable relationship between localisation performance and lesion properties such as area, aspect ratio, boundary contrast and inter-reader agreement. This is the direction with the most scientific payoff, and the constraint here was compute, not method.

Methodological hardening. Multi-seed training should become the default rather than an audit, with seed-level variability reported alongside every test-set interval. Selection noise should be measured by varying the augmentation draw independently of the training seed, and the random control should be run as a permutation distribution rather than as a single draw. Checkpoint selection on a larger or cross-validated split would test the mechanism proposed for the training variability observed here.

Generative-side extensions. Guidance scale remains an untested axis. Generation-seed variability in the maps should be quantified in the same way training-seed variability was. Decomposing the variance signal into edge-driven and pathology-driven components, motivated by the dose–response result, is a more targeted use of compute than enlarging ensembles. Deep ensembles, test-time augmentation and conformal prediction would provide stronger comparators than those available here.

External validity. Cross-dataset transfer to an independent chest radiograph collection would test whether the cross-class pattern is a property of the finding types or of this dataset’s annotation practice. A reader study would address the question no ranking metric answers: whether these maps are informative to the clinicians who would use them.

Appendix A

Supplementary Results

A.1 Per-Arm ROC and Precision–Recall Curves

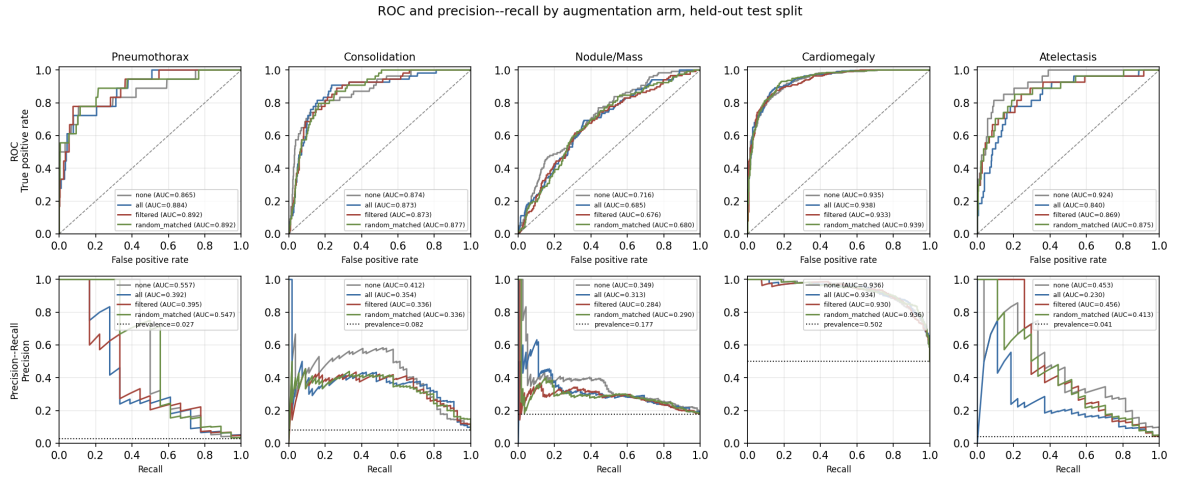


Figure A.1: Per-class ROC and precision–recall curves for each augmentation arm on the held-out test split. Prevalence baselines are marked on the precision–recall panels, without which curves for the rarer classes are not comparable to those for Cardiomegaly.

A.2 Comparative AUC by Arm

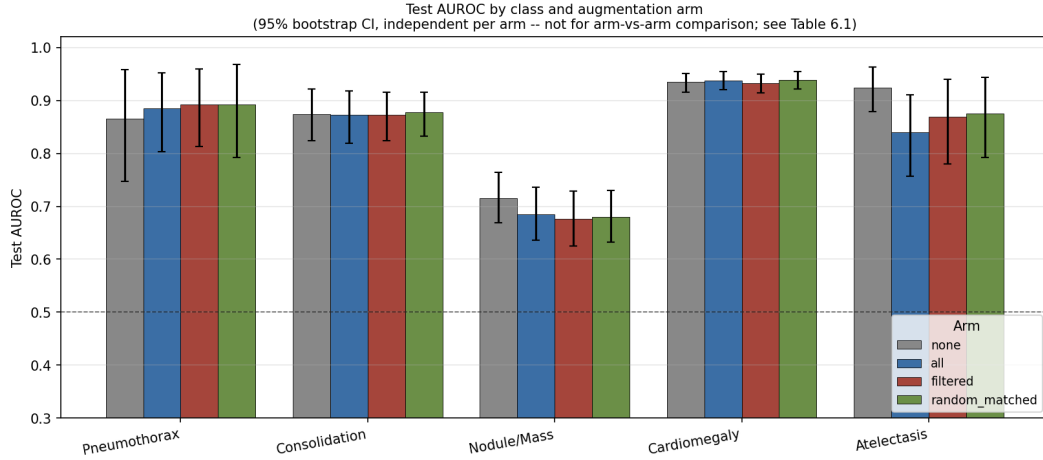


Figure A.2: Test AUROC by class and augmentation arm with bootstrap confidence intervals. These intervals cover test-set sampling variability only; the seed-level variability in Table 6.5 is the larger component and is not shown here.

A.3 Map Resolution Audit

Table A.1: Native spatial resolution of each uncertainty map type as stored, prior to any resampling. All three types were found at native 512×512 , so no resampling confound applies to the comparisons in Section 5.4: the gap between methods is not attributable to coarse maps being upsampled into artificially smooth, high-overlap regions.

Method	Native resolution
Diffusion variance	512×512
MC-Dropout logit variance	512×512
MC-Dropout Grad-CAM variance	512×512

Appendix B

Implementation Details

B.1 Pipeline Stages and Artefacts

The work was executed as a sequence of stages, each producing versioned artefacts consumed by the next. Data preparation and generative model setup produced the image archive and split manifest. Baseline uncertainty estimation established the latent anchoring protocol and the first variance maps. Ensemble generation produced the per-class map sets and the configuration ablation. Comparator generation produced the Monte Carlo dropout and Grad-CAM maps. Downstream training produced the classifier checkpoints for each augmentation arm. Evaluation produced the per-image metric tables and statistical outputs. Replication produced the multi-seed checkpoints and the cross-seed analysis reported in Section 6.4.

B.2 Software Environment

Principal dependencies were PyTorch, Hugging Face `diffusers`, `scikit-learn`, `scipy`, `statsmodels` and `pandas`. Classifier initialisation weights were pinned to a fixed local

Table B.1: Software environment and package versions used for the experiments.

Software / Package	Version
Python	3.14
PyTorch	2.8.2
Hugging Face Diffusers	0.38.0
scikit-learn	1.9.0
SciPy	1.18.0
statsmodels	0.14.5
pandas	3.0.4

copy rather than fetched at run time, so that replication arms share byte-identical initialisation with the runs they replicate.

B.3 Compute Budget

Classifier training took approximately twenty-five minutes per run on the hardware used. The five-seed replication of two arms therefore required ten runs, and each additional filtering threshold required two further runs. Generation dominated total compute, since each class required an ensemble per annotated image.

B.4 Code and Data Availability

Code, configuration files and analysis notebooks are released at the repository listed here.

Gitlab Respository link : <https://git.cs.bham.ac.uk/projects-2025-26/pxc569>

The dataset is available from its original providers under their access conditions and is not redistributed. Generated uncertainty maps, trained checkpoints and per-image metric tables are retained as versioned artefacts.

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